AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

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Team Name: *BRAIN BYTES*

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Problem Statement

Lunar rovers exploring craters, ridges and permanently shadowed regions experience communication blackouts due to terrain-induced line-of-sight obstructions. Existing orbital relays cannot eliminate every local communication dead zone, limiting rover mobility, scientific operations and reliable data transmission.

Problem Understanding

- Who experiences this problem? Lunar rover operators, space agencies, mission-control teams and researchers conducting exploration in challenging lunar terrain.
- What causes the problem? Crater walls, mountains and ridges block direct radio signals. Limited surface infrastructure and intermittent orbital visibility further restrict connectivity.
- How does it affect users? Communication losses interrupt scientific-data transmission, delay command delivery, restrict exploration coverage and increase mission risk.

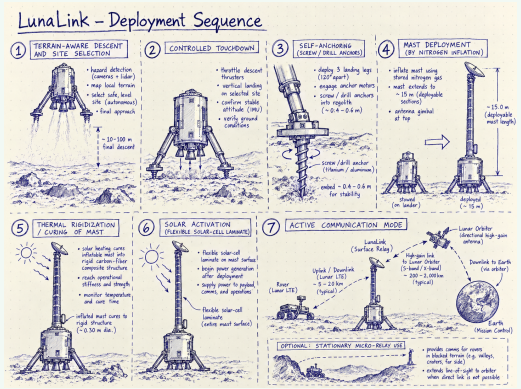
Existing Gaps & Challenges

- Existing gap 1 — Coverage limitations: Orbital relay networks improve regional communication, but local terrain shadows can still prevent reliable rover connectivity.
- Existing gap 2 — Infrastructure limitations: Deployable lunar towers, inflatable structures, solar arrays and anchoring systems have been studied, but their integration into a compact, autonomous relay lander remains insufficiently validated.
- Main unsolved challenge: Developing a stable, energy-efficient, self-deploying communication node that can withstand lunar regolith conditions, thermal extremes, dust and deployment failures.

Our Proposed Solution

- Core idea: LunaLink is an autonomous three-legged lunar lander equipped with a self-anchoring mechanism and a proposed 15 m inflatable, rigidizable carbon-fiber communication mast with integrated flexible solar cells.
- How it works: After landing, the system verifies its stability, anchors into lunar regolith, deploys and rigidizes the mast, and connects nearby rovers to compatible orbital relay satellites.
- Value delivered: Improves local communication coverage, supports data transmission from terrain-obstructed regions and enables scalable communication infrastructure for future lunar exploration.

Technical Approach



- **Autonomous Landing:** Three-legged lander performs controlled touchdown.
- **Self-Anchoring:** Screw anchors secure the lander, with sensors verifying stability.
- **Mast Deployment:** Inflatable carbon-fiber mast extends and rigidizes.
- **Power Generation:** Mast-integrated flexible solar cells charge batteries.
- **Communication:** LTE/4G connects rovers; RF relays data to lunar orbiters.
- **Autonomous Control:** Onboard systems monitor stability, power and connectivity.

Innovation & Uniqueness

- LunaLink integrates an inflatable rigidizable mast, mast-mounted flexible solar cells, autonomous regolith anchoring and surface-to-orbit communication within one compact lander.
- Compact stowage supports transportation, an elevated mast improves local line of sight, and self-anchoring provides a proposed means of structural stabilization.
- An integrated structure–power–communication architecture with automated deployment verification and proposed fault-tolerant operation.]

Feasibility & Scalability

- **Feasibility:**Existing research supports the constituent technologies. Development requires CAD modelling, structural and thermal simulations, regolith-simulant anchoring tests, radio-link analysis and scaled prototype validation.
- **Scalability:** Multiple LunaLink nodes can be positioned at selected terrain locations, with optional stationary micro-relays to expand local rover communication coverage.
- **Risks:** Mast buckling, weak anchoring, solar shadowing, thermal extremes and deployment failures require rigidization testing, anchor verification, power-budget analysis, environmental protection and backup operating modes.

Cost for Product (Optional)

- Development or hardware cost: Estimated 5–10 lakh for a laboratory-scale prototype, including the composite mast, anchoring mechanism, flexible solar panels, communication modules, sensors and testing equipment.
- Hosting or maintenance cost: Estimated 50,000–1 lakh for initial software, monitoring, replacement components and prototype maintenance.
- Estimated total: 5.5–11 lakh for a laboratory-scale prototype, excluding lunar launch, landing, space qualification and mission operations.

Use Cases (compulsorily 3 use cases must be there)

- Use case 1:Crater Exploration: A scientific rover operating behind a crater ridge communicates through an elevated LunaLink node, improving access to mission commands and data transmission.
- Use case 2: Multi-Rover Operations: Multiple lunar exploration rovers connect to a shared LunaLink station for telemetry and scientific-data transfer, supporting coordinated missions.
- Use case 3: Polar Research Missions: Rovers operating in lunar south-pole regions use LunaLink's local communication and store-and-forward capabilities to deliver data during available orbital relay contacts.
  Note: Additional slides may be included only for this section.

Impacts and Benefits

- User benefits: Improved local rover connectivity, more flexible exploration routes and greater access to scientific-data transmission.
- Scientific and economic impact: Supports lunar exploration research and may reduce duplicated communication infrastructure across future missions.
- Expected measurable outcomes: Improvements in line-of-sight coverage, communication availability, successful data delivery, deployment reliability and energy autonomy, to be assessed through modelling and testing.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 9	Target 9.5	Industry, Innovation and Infrastructure	Advances engineering research and the development of innovative communication infrastructure
SDG 17	Target 17.6	Partnerships for the Goals	Potential support for future interoperable technology development and scientific collaboration

Individual Member Contribution

Member	Contribution or responsibility
Tharunima M	TEAM LEADER
Jagat S R	PRESENTER
Sanjay Krishnan V	USE CASE DESIGN
Vishwaja S	IDEA CURATION

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